

Connector for ODK - User Manual

Version 1 - Administrator (includes KeSMIS Import)

Plugin version 2.0

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Homepage: <https://github.com/fnmutua/connector-for-ODK>

1. Introduction

Connector for ODK is a QGIS plugin with four tools:

Tool	Purpose
Get Data	Download ODK Central form submissions and load them as map layers
Split Layer	Split a vector layer into separate layers by attribute value
QA/QC	Run quality checks on File Geodatabase layers and export issue reports
Import	Upload QGIS vector features to a KeSMIS server

Each dialog includes a collapsible Help panel. The QA/QC dialog opens with help visible; other tools show << Show Help to open it.

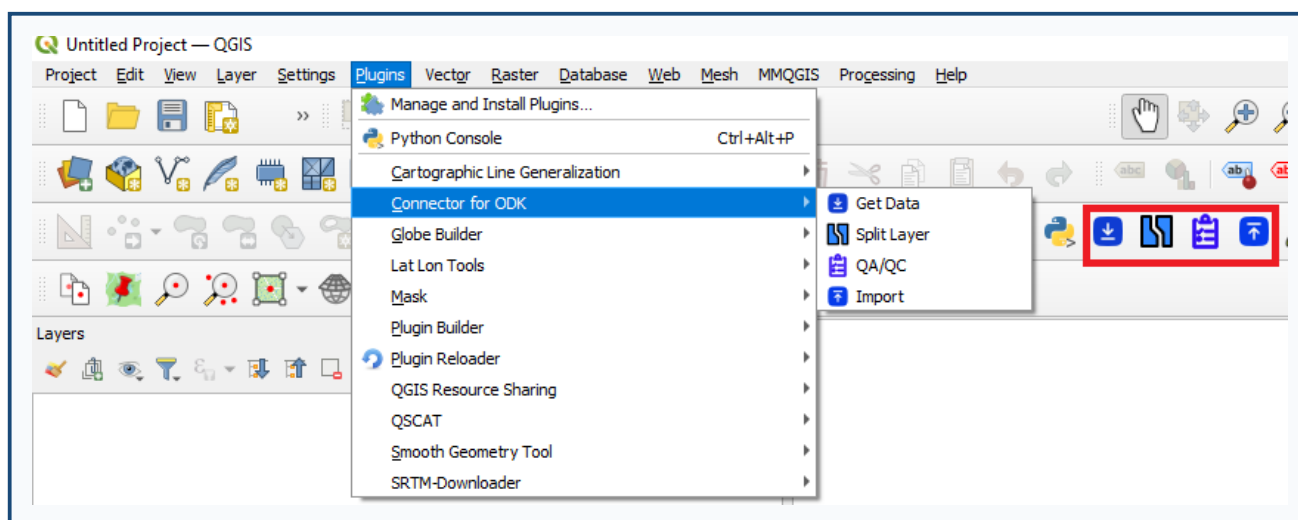


Figure 1 - QGIS menu and toolbar with Get Data, Split Layer, QA/QC, and Import

2. Prerequisites

2.1 QGIS

- QGIS 3.0 or later (desktop)
- An internet connection for ODK Central, KeSMIS, and optional package installation

2.2 Python packages

The plugin installs missing packages automatically the first time it loads. If that fails, install dependencies

manually using OSGeo4W Shell from your QGIS installation (Windows).

Open OSGeo4W Shell

1. Close QGIS.
2. Open OSGeo4W Shell from your QGIS installation folder - for example C:\Program Files\QGIS 3.x\bin\OSGeo4W.bat - or from the Windows Start menu under your QGIS install (e.g. QGIS Desktop -> OSGeo4W Shell).

Use the OSGeo4W Shell that ships with the same QGIS version you use. Do not use a separate OSGeo4W install, or packages may install into the wrong Python environment.

Install required packages

Run this command in that shell:

```
python -m pip install numpy pandas geopandas fiona shapely pyproj fpdf2 requests fuzzywuzzy
rapidfuzz shortuuid openpyxl xlswriter
```

Required packages:

Package	Used for
numpy	Numerical processing
pandas	Tables and spreadsheets
geopandas	Spatial data handling
fiona	Reading/writing geospatial files
shapely	Geometry operations
pyproj	Coordinate reference systems
fpdf2	QA/QC PDF reports (import fpdf)
requests	ODK Central and KeSMIS API calls
fuzzywuzzy	Fuzzy field and attribute matching
rapidfuzz	Fast field matching (Import)
shortuuid	Unique ID generation (Import)
openpyxl	Reading dictionary.xlsx (QA/QC)
xlswriter	Writing QA/QC Excel outputs

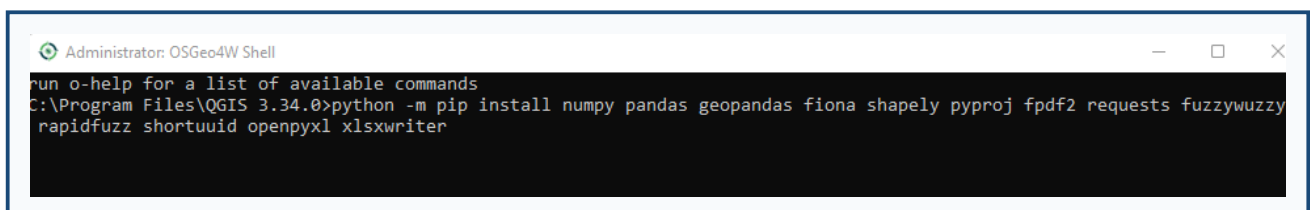


Figure 2 - Installing Python packages in OSGeo4W Shell from your QGIS installation folder

Troubleshooting: fixing a missing package error

If QGIS shows a yellow bar such as `Couldn't load plugin 'connect_odk'...` or the Message Log reports `ModuleNotFoundError: No module named '...',` a required Python package is not installed in the QGIS Python environment.

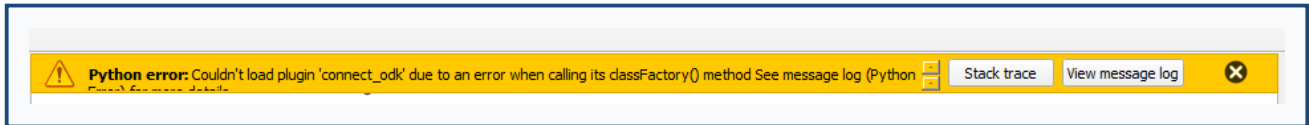


Figure 3 - Plugin load error when a required Python package is missing

Click View message log (or open View -> Panels -> Log Messages -> Python Error) to see which package is missing.

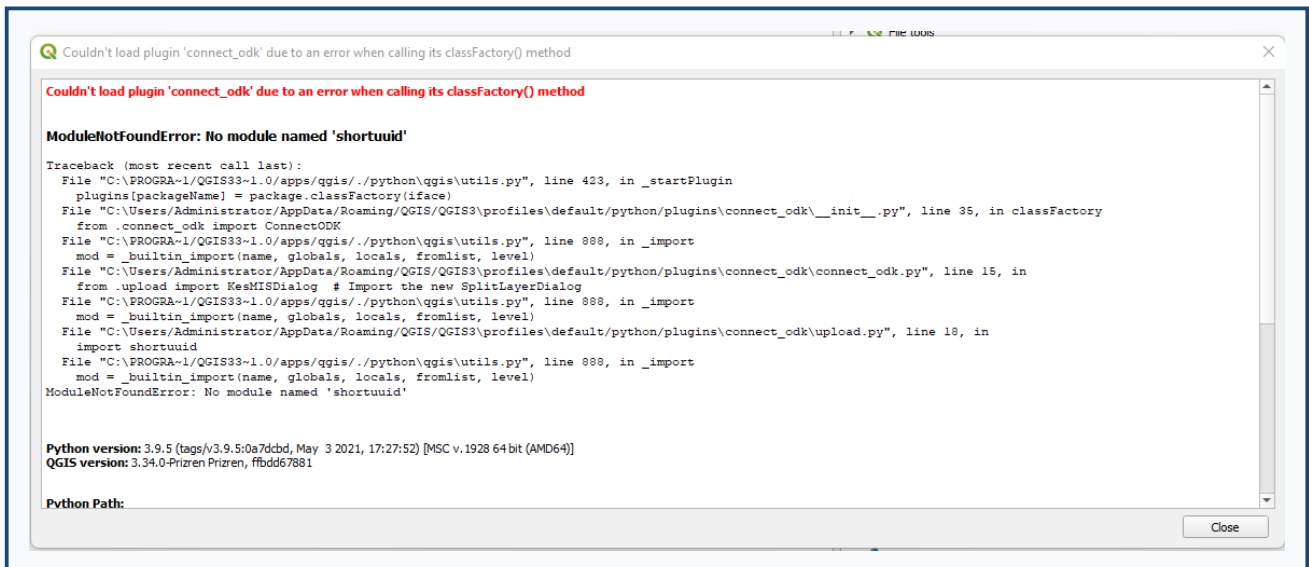


Figure 4 - Message log showing the missing package (example: shortuuid)

How to fix it

1. Note the package name in the error (the name inside quotes after No module named).
2. Close QGIS completely.
3. Open OSGeo4W Shell from your QGIS installation folder (see above).
4. Install the missing package. You can install all required packages (recommended):

```
python -m pip install numpy pandas geopandas fiona shapely pyproj fpdf2 requests fuzzywuzzy
rapidfuzz shortuuid openpyxl xlswriter
```

Or install only the missing package (replace shortuuid with the name from your error):

```
python -m pip install shortuuid
```

5. Restart QGIS. Connector for ODK should load without the error.

2.3 Data and access

Depending on which tools you use, you may also need:

- Get Data - ODK Central URL, username, and password
- Split Layer - Vector layers already loaded in the QGIS project
- QA/QC - An ESRI File Geodatabase (.gdb folder)
- Import - KeSMIS server URL, username, and password; vector layers loaded in QGIS

3. Installing the Plugin

Option A - QGIS Plugin Repository (recommended)

1. Open QGIS.
2. Go to Plugins -> Manage and Install Plugins.
3. Search for Connector for ODK.
4. Click Install Plugin.
5. Restart QGIS if prompted.

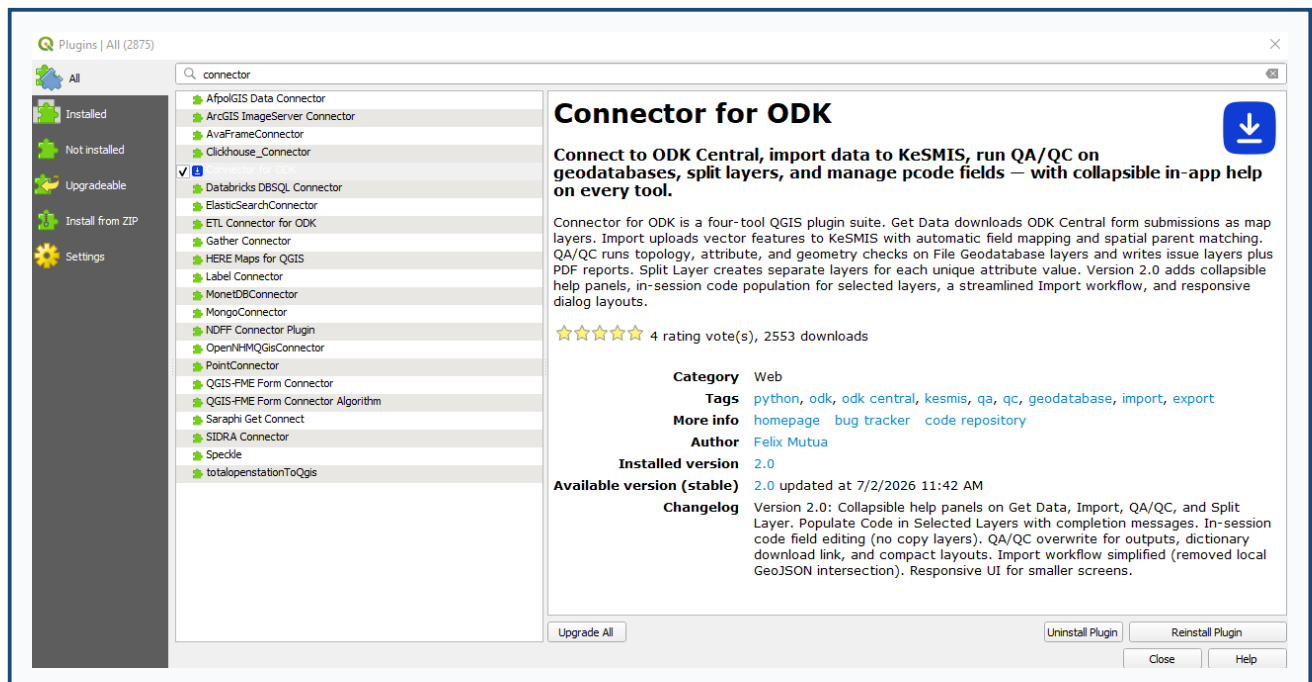


Figure 5 - Plugin Manager with Connector for ODK selected

Option B - Install from ZIP

1. Download connect_odk.zip (version 2.0).
2. In QGIS, go to Plugins -> Manage and Install Plugins.
3. Open the Install from ZIP tab.
4. Select the ZIP file and click Install Plugin.
5. Restart QGIS if prompted.

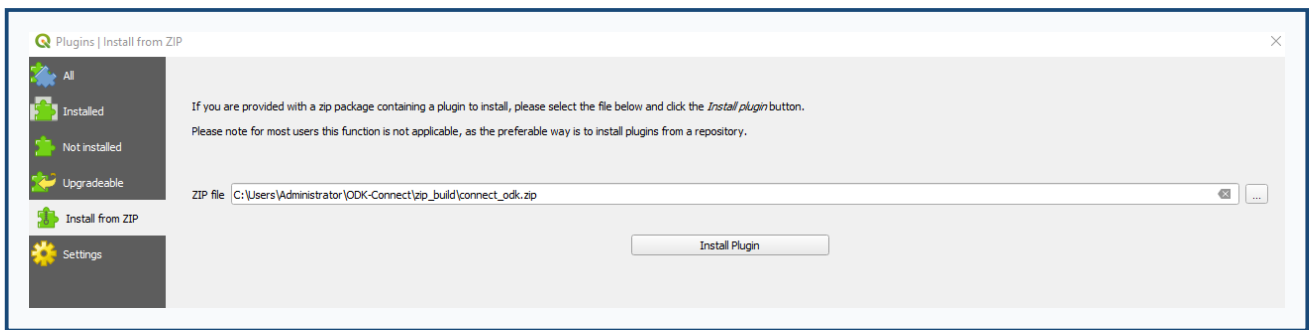


Figure 6 - Installing the plugin from a ZIP file

Verify installation

After QGIS restarts, you should see:

- Menu: Plugins -> Connector for ODK
- Toolbar icons for Get Data, Split Layer, QA/QC, and Import

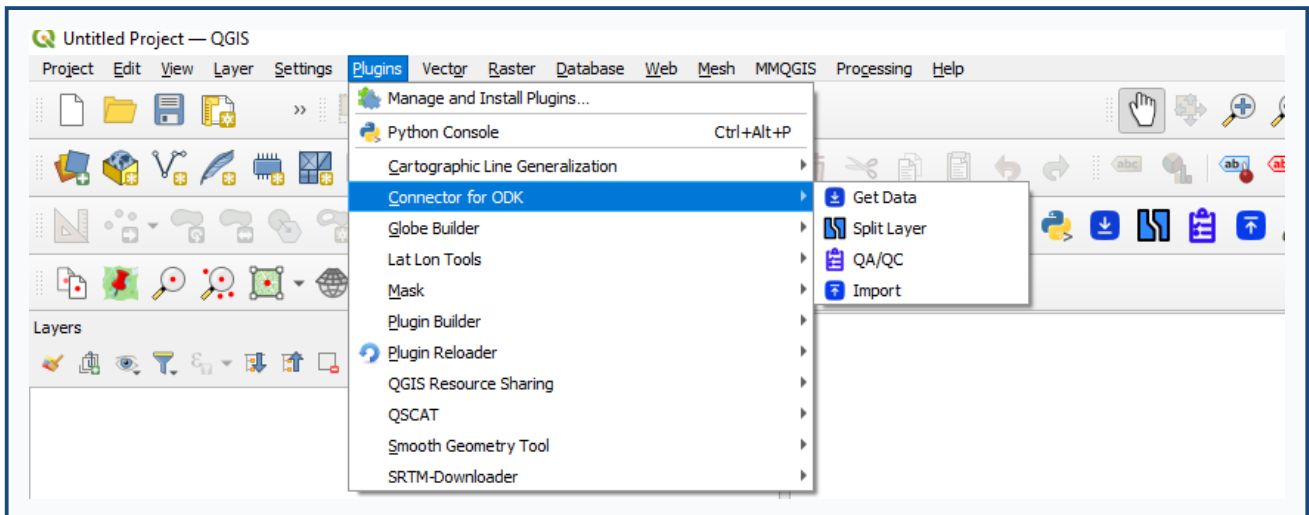


Figure 7 - Plugins menu with all four tools listed

4. Get Data (ODK Central)

Download ODK form submissions and add them to your map.

Figure 8 - Get Data dialog with ODK Central login

Steps

1. Open Plugins -> Connector for ODK -> Get Data.
2. Enter your ODK Central URL, username, and password.
3. Click Login to load projects and forms.
4. Select a project and form.
5. Click Process Form to fetch submissions and add a GeoJSON layer to the map.
6. Optionally click Get CSV to export the data as a spreadsheet.

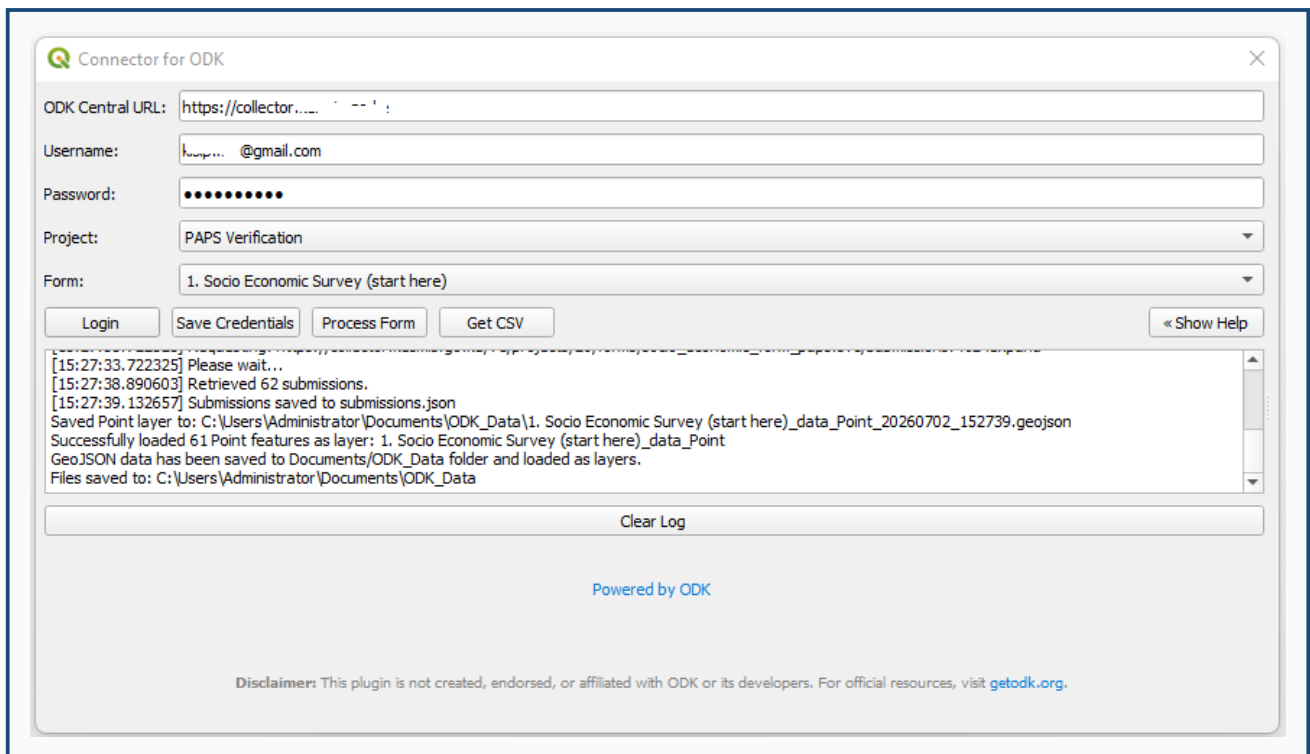


Figure 9 - Project and form selected before processing

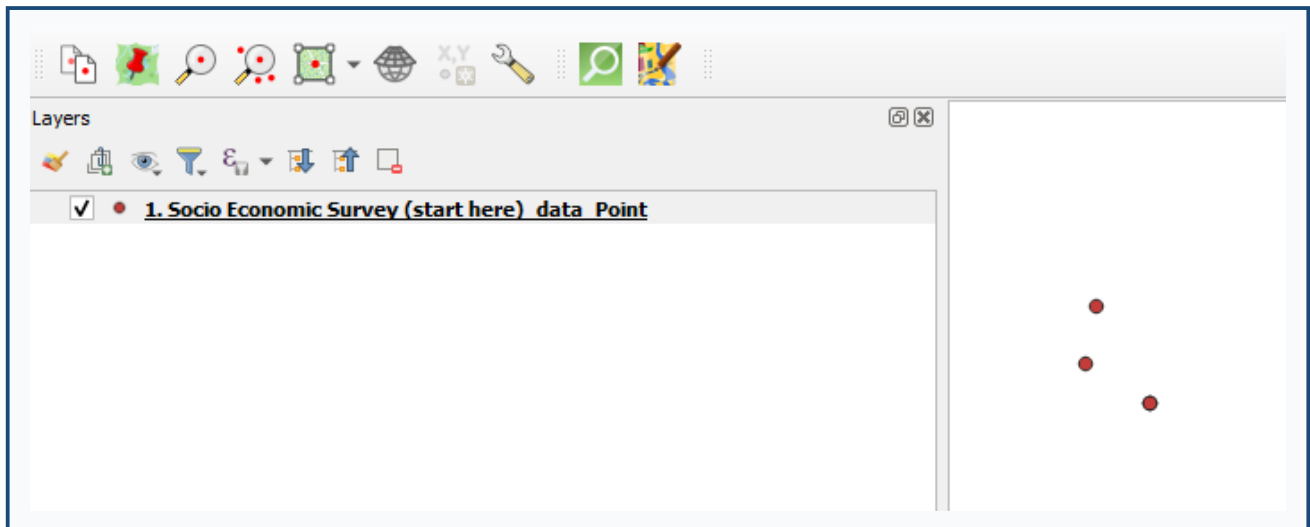


Figure 10 - Submissions loaded as a layer on the QGIS map

Notes

- Use Save Credentials to store your URL and login for next time.
- Output is in EPSG:4326 (WGS 84).
- A submissions.json file is written to the working folder.
- Check the Log panel at the bottom of the dialog for progress and errors.

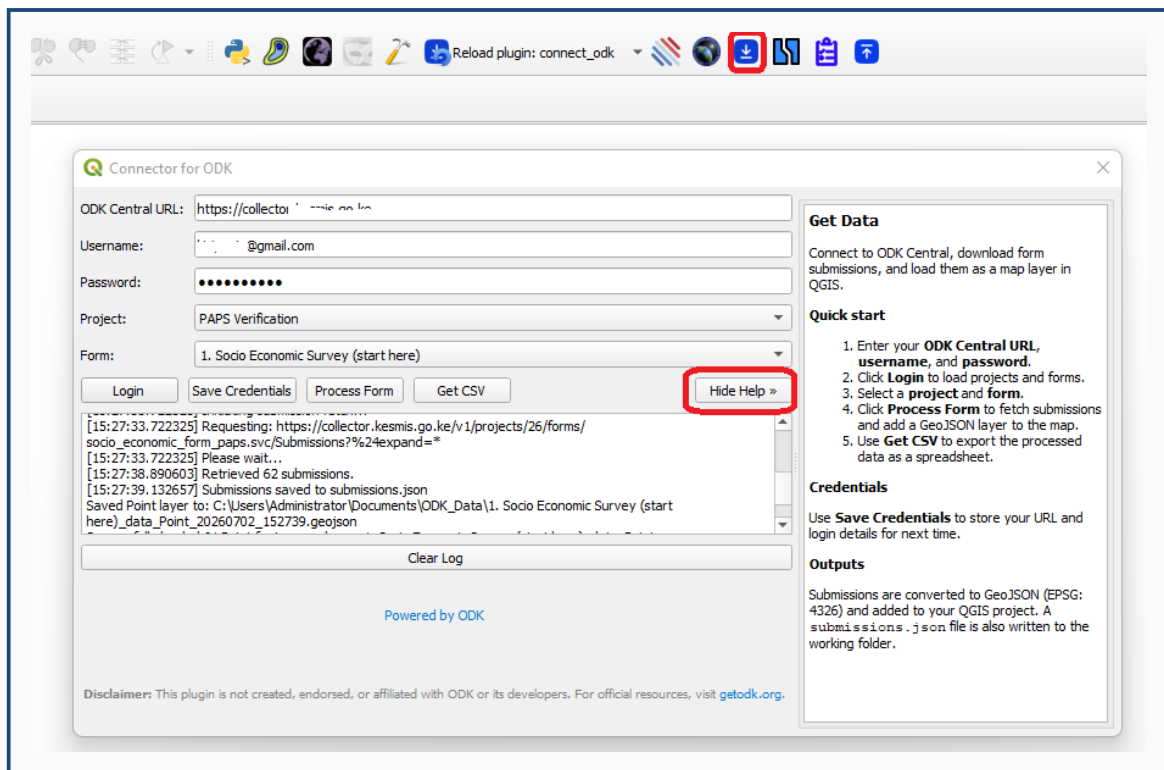


Figure 11 - Collapsible help panel in Get Data

5. Split Layer

Create separate in-memory layers for each unique value in a chosen attribute field.

Steps

1. Load the source vector layer in QGIS.
2. Open Plugins -> Connector for ODK -> Split Layer.
3. Choose a Layer from the project.
4. Choose the Field to split on.
5. Click Split Layer.

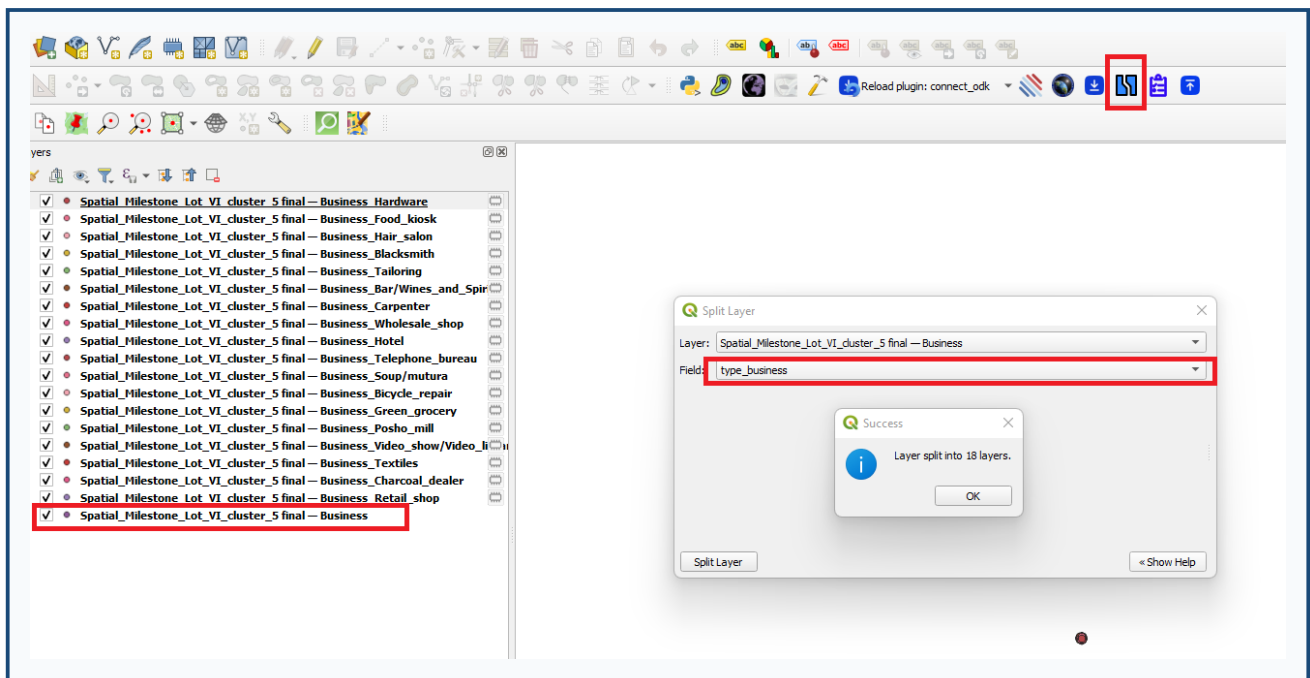


Figure 12 - Split Layer dialog

Result

After splitting:

- One new layer is created per unique non-null value.
- Layers are named {layer}_{value}.
- Empty fields are dropped from each split layer.
- Geometry and CRS are copied from the source.

6. QA/QC

Run quality checks on File Geodatabase layers and export issue layers, spreadsheets, and a PDF summary.

Steps

1. Open Plugins -> Connector for ODK -> QA/QC. The dialog opens with the help panel visible.
2. From Quick start step 2 in the help panel, download the template geodatabase and dictionary.xlsx for reference when aligning your data (see below).
3. Click Select GeoDatabase and choose the folder containing your .gdb. Use the template to structure new submissions.
4. Click Select Output Folder for reports and issue layers.
5. Adjust parameters if needed (angle and length thresholds).
6. Under Select Layers, tick the layers to check, or use Select All. The layer list scrolls inside a fixed panel so Run All Checks stays visible.
7. Click Run All Checks.
8. When finished, use the PDF Report and Open Output Folder links below the log.

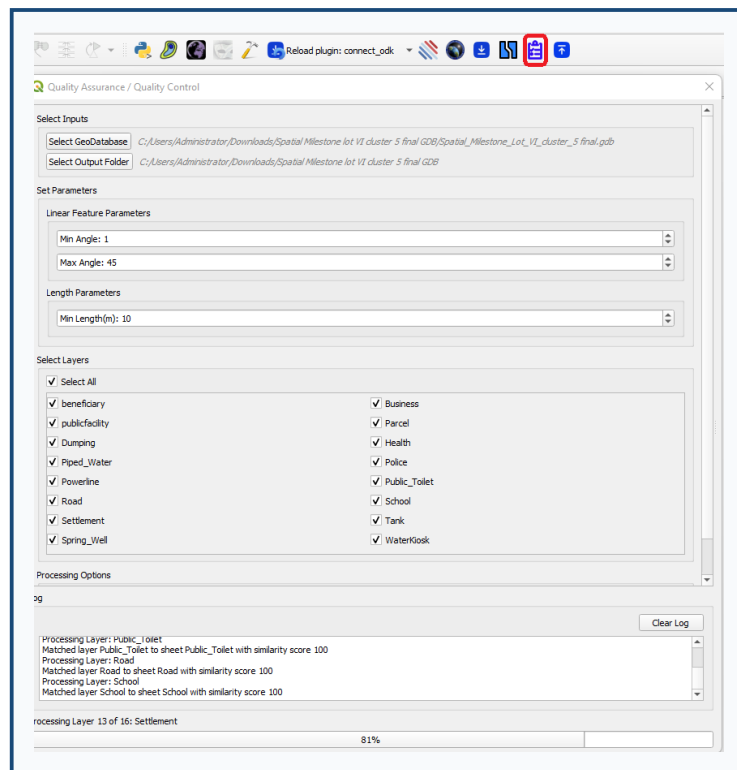


Figure 13 - QA/QC interface

Checks performed

Check	Description
Duplicate geometries	Features with identical geometry
Duplicate attributes	Rows with identical non-geometry fields
Overlapping polygons	Polygon pairs sharing area above 0.01 m2
Line issues	Sharp turns and self-intersections
Short lines	Line features shorter than the minimum length
Attribute issues	Fields validated against dictionary.xlsx

Parameters (defaults)

Parameter	Default	Purpose
Min Angle	1 deg	Lower bound for flagged turn angles
Max Angle	45 deg	Upper bound for flagged turn angles
Min Length	10 m	Flag lines shorter than this

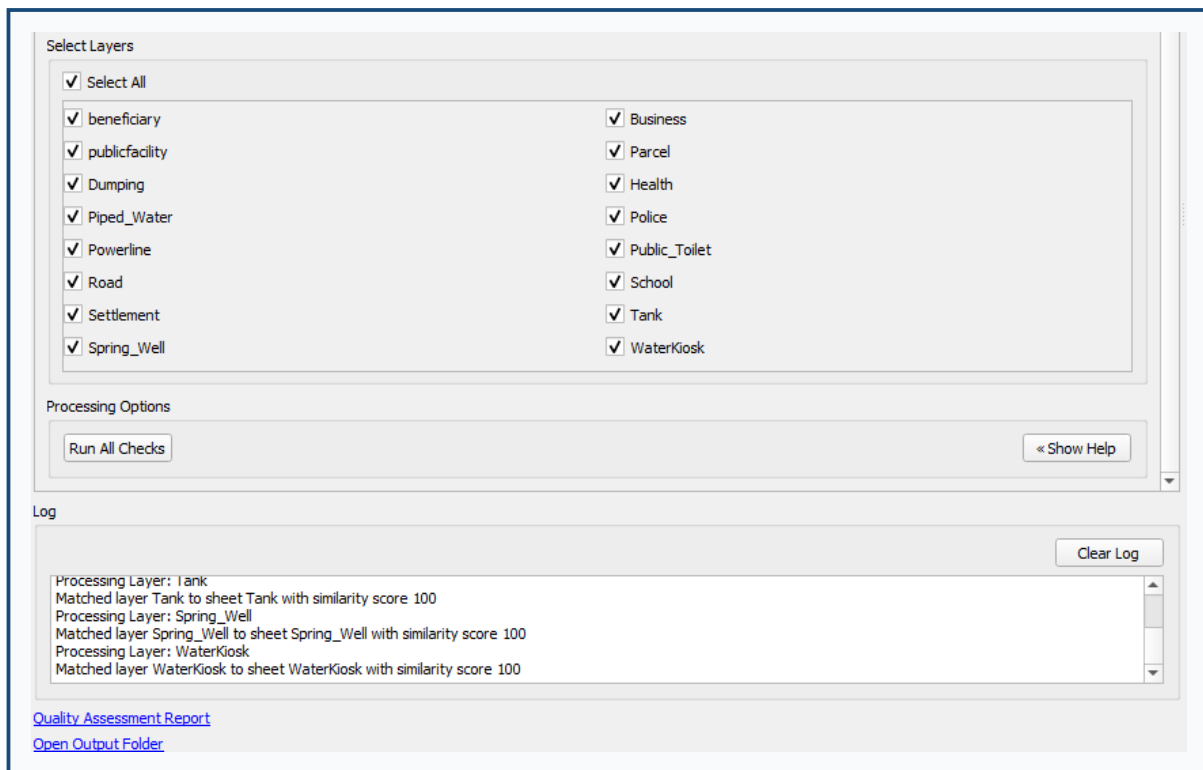


Figure 14 - Completed run with report and output folder links

Template geodatabase and attribute dictionary

The template geodatabase and attribute dictionary are reference materials for packaging and QA/QC. They describe the expected layer names, fields, types, and geometry so your .gdb aligns with the real submission structure. Download copies from Quick start step 2 in the QA/QC help panel.

1. Download template geodatabase - shows the expected layer layout. Use it as a guide when building or checking your geodatabase.
2. Download dictionary.xlsx - lists the field names, types, and geometry for each layer. Open the How to sheet for column definitions; every other sheet describes one layer.
3. Using both together - consultants use the template and dictionary as reference when packaging data so layers and attributes match what QA/QC expects.
4. Help panel downloads - links open a save dialog without clearing the help text.

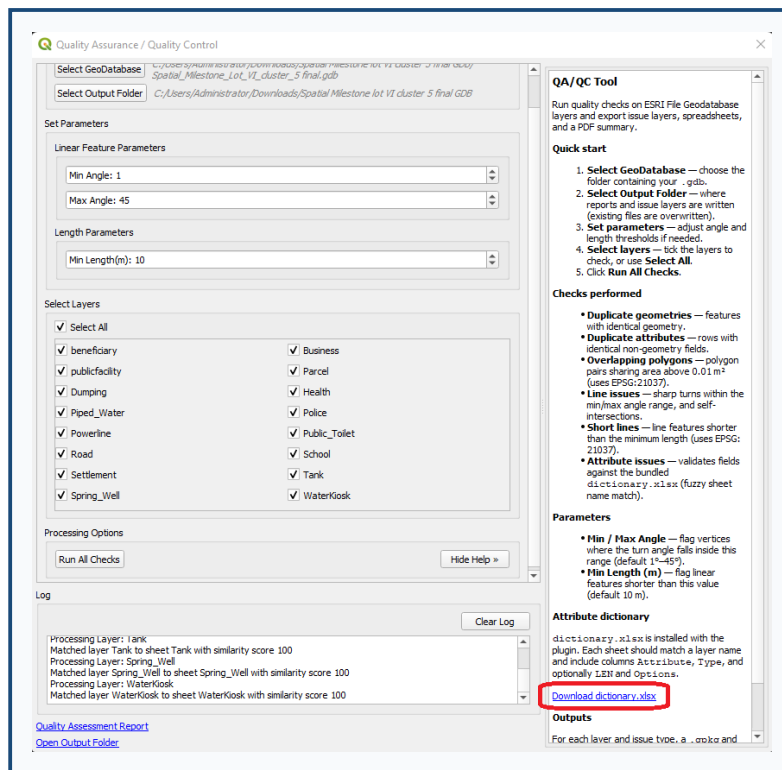


Figure 15 - Help panel showing template geodatabase and dictionary download links

Outputs

For each layer and issue type, .gpkg and .xlsx files are written to the output folder. A summary PDF (database_summary_report.pdf) is also created. Existing output files are overwritten on re-run.

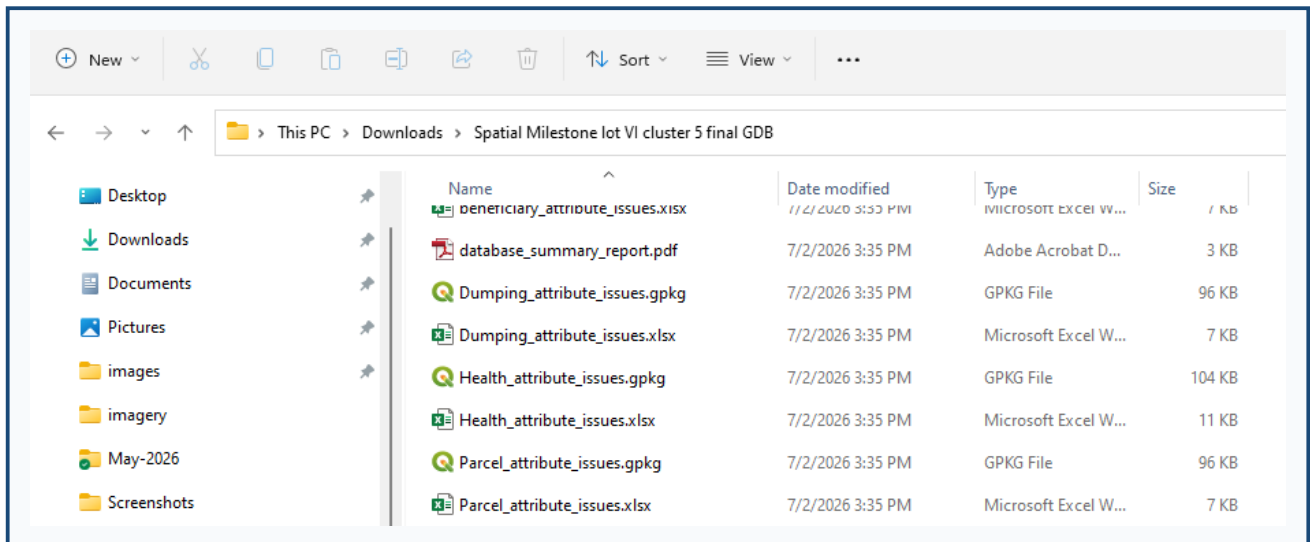


Figure 16 - Example QA/QC output files in the output folder

7. Import (KeSMIS)

Upload QGIS vector layer features to a KeSMIS server with automatic field mapping and parent-entity assignment.

KeSMIS login

When you open Import, a KeSMIS Login popup appears first. You must enter the server URL, username, and password and click Login before the main import dialog opens. Click Cancel to abort.

- Save credentials stores your URL, username, password, and session token for next time.
- If a saved token is still valid, the popup shows Logging in... with a progress bar and opens the import dialog automatically.
- If the token has expired, the full login form is shown so you can sign in again.

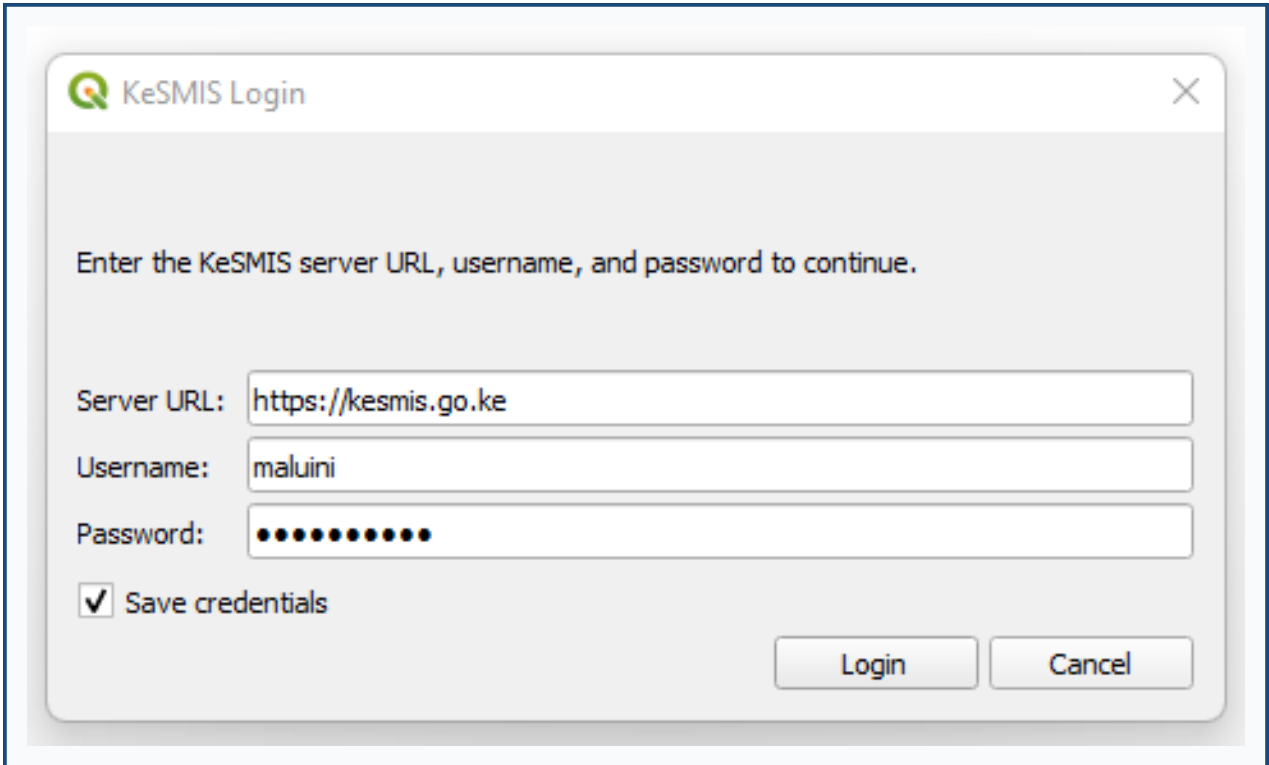
A screenshot of the KeSMIS Login popup dialog. The dialog has a title bar with the KeSMIS logo and the text "KeSMIS Login". Inside the dialog, there is a message: "Enter the KeSMIS server URL, username, and password to continue." Below this message are three input fields: "Server URL:" with the value "https://kesmis.go.ke", "Username:" with the value "maluini", and "Password:" with a masked password represented by ten dots. Below the password field is a checkbox labeled "Save credentials" which is checked. At the bottom right of the dialog are two buttons: "Login" and "Cancel".

Figure 17 - KeSMIS Login popup

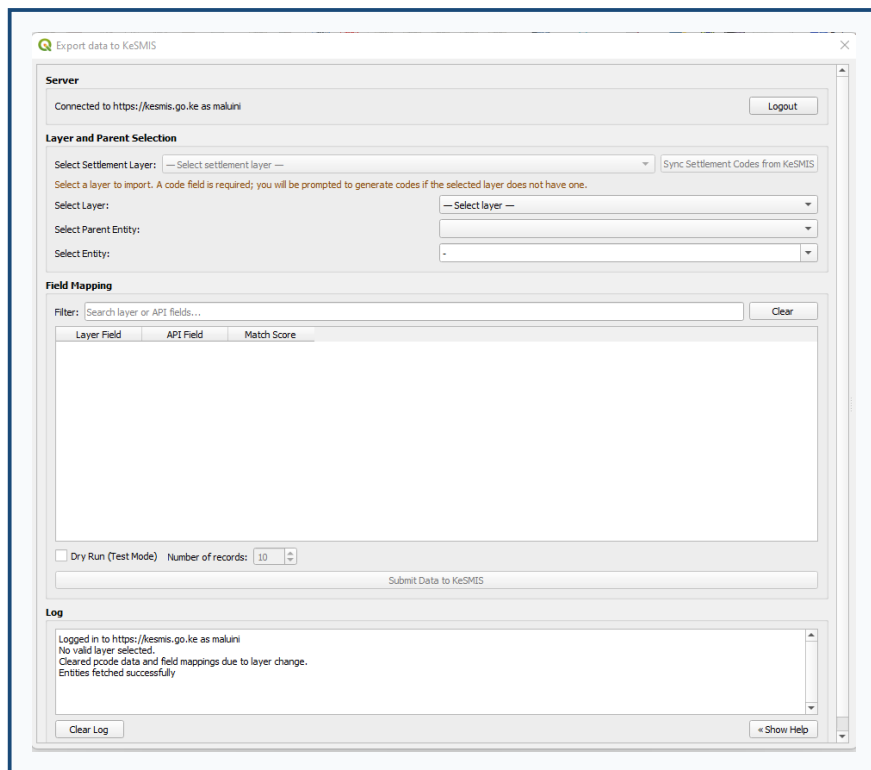


Figure 18 - Import dialog after login (Server status, Logout, layer selection, field mapping)

Overview

After login, the import dialog is organized top to bottom:

1. Server - shows Connected to {url} as {username}. Click Logout to sign out and close the dialog (clears the saved session token).
2. Settlement codes (recommended first) - sync official KeSMIS codes into settlement layers before upload.
3. Select Layer - choose the vector layer to export (starts empty until you pick one).
4. Select Parent Entity - choose settlement or ward for spatial parent matching.
5. Select Entity - choose the API model to submit to; field matching runs after you confirm your choice.
6. Field Mapping - review or adjust automatic field matches, then submit.

Plugin dialogs stay scoped to the QGIS window and do not float over other applications.

Settlement codes (recommended first step)

If your upload layer is a settlement layer (layer name contains settlement, excluding reference layers ending in Boundaries), use KeSMIS codes - do not generate random UUIDs. Each settlement feature needs the same code as KeSMIS so imports update existing records instead of creating duplicate settlements on the server.

1. Open Plugins -> Connector for ODK -> Import and complete the KeSMIS Login popup.
2. Under Select Settlement Layer, choose your settlement layer from the dropdown (nothing is pre-selected).
3. Click Sync Settlement Codes from KeSMIS. This matches your local features to official KeSMIS settlements by geometry and copies the server code onto your layer. Using those codes on import prevents duplicate settlement records when the same place already exists in KeSMIS.

4. Review the Review Settlement Matches table:

- One KeSMIS match - the best overlap is selected automatically.
- Multiple KeSMIS matches - choose the correct settlement from the dropdown for that row.
- No intersection - a new short code is generated and shown before you confirm; you can proceed with that code.

5. Click Transfer Codes to apply codes to the layer.

After a successful sync, features are marked in the `kesmis_sync` field and syncing is disabled for that layer to protect official codes. To re-sync, remove the `kesmis_sync` field from the layer first.

Upload workflow

1. Complete KeSMIS Login when the popup appears (see above).
2. Select Layer - choose the QGIS vector layer to export. The dropdown starts on - Select layer -; pick a layer explicitly.
3. Code field check - when you select a layer:
 - If the layer already has a code field, loading continues normally.
 - If the layer has no code field (and is not a settlement layer), you are prompted to generate unique codes. Choose Yes to add/fill codes, or No to cancel - import cannot continue without a code field.
 - If the selected layer looks like a settlement layer, you are directed to use Sync Settlement Codes from KeSMIS instead of generating random codes.
4. Select Parent Entity - choose settlement or ward. When prompted, download fresh parent boundaries from the server (recommended) or use a cached local GeoJSON file.
5. Select Entity - search or browse the entity list, then click your choice (or press Enter). Field matching starts only after you confirm the entity - browsing the list does not start matching early.
6. Review the Field Mapping table. Use the filter box to search rows. In each API Field cell:
 - Open the dropdown to see the full list of API fields (even when a match already exists).
 - Type to filter the list, or type a custom API field name if needed.
 - Choose - to clear a mapping.
7. Dry run (optional) - enable Dry Run (Test Mode) and set how many records to validate. Click Validate on KeSMIS to send a test batch with `dryRun: true`. Nothing is saved; the summary shows what would be inserted, updated, or failed, and any errors are listed in the dialog.
8. Submit - disable dry run and click Submit Data to KeSMIS for the full upload.

To sign out, click Logout in the Server section. The import dialog closes and the saved session token is cleared; open Import again to log in.

Dry run

Enable Dry Run (Test Mode) to validate a limited number of records on the server without saving anything. Set the record count in the spin box next to the checkbox.

- The plugin sends `dryRun: true` to the KeSMIS import API.
- The server returns the same style of counts as a real import (`insertedCount`, `updatedCount`, `failedCount`).

- If any records fail validation, errors are shown in the Dry Run Complete dialog as well as in the log.
- When you are satisfied with the dry run, turn off dry run and submit for real.

8. Tips and Troubleshooting

Issue	What to try
Plugin does not appear after install	Restart QGIS. Check Plugins -> Manage and Install Plugins -> Installed that Connector for ODK is enabled.
Package install fails	See Fixing a missing package error in Section 2.2. Close QGIS, open OSGeo4W Shell from your QGIS installation folder, install the missing package (or all packages), then restart QGIS.
ODK / KeSMIS login fails	Confirm URL, username, and password in the KeSMIS Login popup. Check network access to the server.
KeSMIS session expired	Open Import again and sign in. If auto-login fails, enter credentials manually. Use Logout to clear a stale session.
Import missing code field	When you select a layer, choose Yes when prompted to generate codes. Settlement layers must use Sync Settlement Codes from KeSMIS, not random UUIDs.
Settlement codes wrong or missing	Use Select Settlement Layer and Sync Settlement Codes from KeSMIS before upload. Pick the correct KeSMIS match when several overlap.
Field matching starts too early	Finish selecting an entity by clicking it in the list (or pressing Enter). Browsing with arrow keys alone does not run matching.
API field dropdown looks empty	Clear the search text in the cell, or click the dropdown arrow to reopen the full list. Matched fields still show all options when opened.
Dry run still saves data	Ensure the KeSMIS server supports dryRun: true on /api/v1/data/import/upsert. The plugin sends dry run only when Dry Run is checked.
QA/QC attribute check skipped	Ensure dictionary.xlsx has a sheet matching the layer name.
No layers in a dropdown	Load vector layers into the QGIS project first. Layer and settlement dropdowns start empty until you choose one.

For updates, bug reports, and source code:

- Tracker: <https://github.com/fnmutua/connector-for-ODK>
- Repository: <https://github.com/fnmutua/ODK-Connect>